

3(a)	A antibiotic for penicillin throughout <i>three from:</i> 1 (penicillin) weakens / AW, the <u>cell wall</u> ; I punches holes / holes made 2 (penicillin) acts, on growing cells / when cell wall being synthesised (during growth) ; 3 inhibits / binds to / AW, enzymes / transpeptidases (for cross linkage formation) ; I ref. to synthesis of peptidoglycan 4 prevents formation of cross, links / linkages (between, peptidoglycan / murein, molecules) / AW ; <i>suggestions why antibiotic is less effective on Gram negative</i> 5 outer membrane, prevents / interferes with / protects from / AW, entry (of penicillin) ; A idea of, more difficult / further, to reach peptidoglycan layer 6 proteins in outer membrane may pump out antibiotic ; A presence of efflux pumps 7 enzymes may be present (in periplasm) to degrade antibiotic / AW ; 8 suggestion that antibiotic cannot cross hydrophobic region of (outer) membrane ; 9 AVP ; e.g. proportionately, less / lower concentration of, penicillin reaches murein for, enzyme / transpeptidase, inhibition	3
3(b)(i)	ATP not required ; A (metabolic) energy not required	1

Question	Answer	Marks
3(b)(ii)	<i>four from:</i> 1 channel protein ; A pore protein / has a pore / has a channel 2 channel / pore, can form from polypeptides or (protein) has quaternary structure ; 3 hydrophilic R-groups, on amino acids lining channel / face inwards (towards channel) ; A hydrophilic, lining / channel A water-filled, channel / AW A idea that passage is through hydrophilic region of protein 4 allows facilitated <u>diffusion</u> ; A diffusion alone if in context of through the protein but R if via phospholipid bilayer 5 increases permeability for, movement of water / osmosis ; I faster 6 no (specific) binding sites / (channel) not specific / not selective / allows more than one type of substance through / AW ; 7 globular ; 8 AVP ; e.g. ref. to hydrophobic part of protein, faces / interacts with, hydrophobic, region / core / fatty acid tails ref. to hydrophilic parts of, protein / polypeptide, extend into, external environment / periplasm / aqueous regions / AW	4

Question	Answer	Marks
3(b)(iii)	<i>two from:</i> translation, cannot / does not, begin / occur or polypeptide / protein, synthesis, cannot occur / decreases ; I ompF not made A chain of amino acids R transcription does not occur so translation does not occur R if in context of mutation or enzyme inhibition mRNA cannot attach to, ribosome / small subunit / ribosomal subunit ; (first) tRNA (with UAC anticodon) / tRNA carrying met, cannot bind (to START codon) ;	2
3(c)(i)	(composed of) many / chain of / polymer of / AW, monosaccharides / sugar monomers / sugar units ; A glucose, molecules / residues A more than two / many sugars further detail ; e.g. carbohydrate may be, branched / unbranched <i>in context of polysaccharide</i> macromolecule <i>in context of polysaccharide</i> glycosidic bonds <i>in context of between sugar monomers</i> (sugar monomers) joined by condensation reactions	2

Question	Answer	Marks
3(c)(ii)	<i>three from:</i> 1 <u>specificity</u> / <u>specific</u> (in correct context) ; 2 (B / T, -) lymphocytes have receptors complementary to antigen ; A immunoglobulins / antibody <i>as receptors for B-lymphocyte</i> A surface molecules <i>as receptors for T-lymphocytes</i> 3 (different) antigens, stimulate / activate / AW, (different) B-lymphocytes / T-lymphocytes ; A antigens stimulate an immune response 4 <i>idea that</i> different antibodies, synthesised / produced / AW, for different (O) antigens / O polysaccharides / lipopolysaccharides ; 5 memory cells will, not respond to different antigen / only respond to same antigen / AW ; 6 different O-antigens can, be composed of different sugars ; A can have different shapes	3